





CIRP - Project Planning & Management

Customer Intelligence & Retention Platform

DEPI • AI & Machine Learning Track • Round 5

Graduation Project

Project	Customer Churn Prediction & Segmentation Platform (CIRP)
Track	DEPI - AI & Machine Learning
Project Leader	Mohamed Atef Mohamed Abdo
Repository	https://github.com/nhahub/NHA-5-011
Version / Date	v1.0 - October 2026

1. Project Plan

1.1 Timeline (Gantt chart)

The project runs in phases from planning to the final presentation (Figure 1): planning documents (30 Oct), system analysis & design (6 Nov), implementation (30 Nov) and the final presentation (4 Dec).

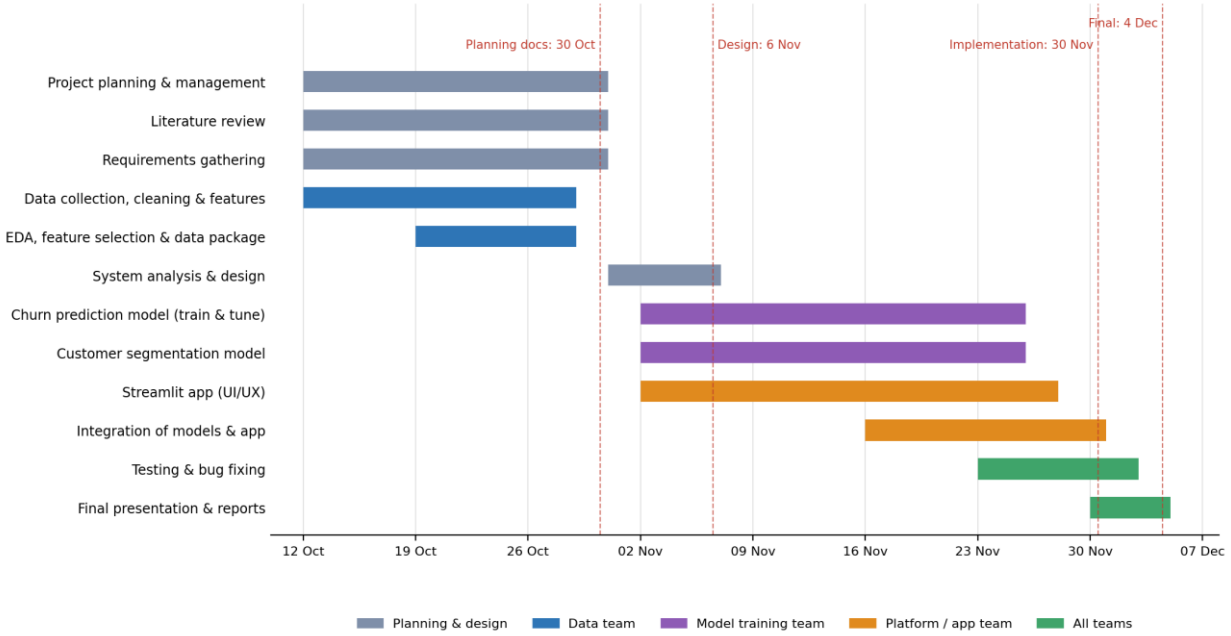


Figure 1: Project Gantt chart

1.2 Phases

Phase	Dates (2026)	Status
Project planning & management	2 - 16 Oct	Done
Literature review	9 - 16 Oct	Done
Requirements gathering	9 - 16 Oct	Done
Data collection, cleaning, features, EDA	12 - 28 Oct	Done
System analysis & design	9 Oct - 6 Nov	In progress
Implementation: models, app, integration	16 Oct - 30 Nov	In progress
Testing & bug fixing	23 Nov - 2 Dec	Planned
Final presentation & reports	16 Oct - 4 Dec	Planned

1.3 Milestones & Deliverables

ID	Milestone / deliverable	Completion criteria
M0	Data package delivered	Cleaned dataset, train/val/test, scaled files, preprocessor, data dictionary and EDA report shared with the teams.
M1	Planning documents submitted	Project Planning & Management, Literature Review and Requirements Gathering uploaded to the LMS (text + GitHub link).

ID	Milestone / deliverable	Completion criteria
M2	System analysis & design complete	Architecture, app wireframes and model-app interface contract agreed and documented.
M3	Models trained and evaluated	Final churn and segmentation models saved, with the evaluation report and the segment profiles; test set used once.
M4	Platform integrated and implementation delivered	Streamlit app runs locally from the repository with the saved models end to end; source code and execution instructions on GitHub.
M5	Final presentation, testing and reports	Test report (2 Dec), final presentation and final report delivered.

2. Task Assignment & Roles

2.1 Team roles

Member	Team / main focus	Responsibilities
Mohamed Atef Mohamed Abdo	Data & Preprocess - data source; Project Leader	Coordinates the team and deadlines, submits project documents, owns the data source, EDA and documentation.
Mohamed Ahmed Ebrahim Mohamed	Data & Preprocess - preprocessing	Cleaning, scaling, feature selection and the reusable preprocessing used by the app.
Omar Diaa El-Sayed	Model Training - segmentation	K-Means / DBSCAN, choosing the number of segments, segment profiles and business names.
Sohaib Mohamed Mohamed Mustafa	Model Training - prediction	Churn classifiers, tuning, threshold selection, explanations and evaluation.
Mohamed Hamada Omar	Application - UI/UX	App design, pages, charts and user experience in Streamlit.
Hussein El-Desouky Ibrahim Othman	Application - integration	Loading models, applying preprocessing, input validation, batch scoring and end-to-end testing.

3. Risk Assessment & Mitigation Plan

ID	Risk	Mitigation
R1	Data leakage inflates the metrics.	Time-based split; scaler and feature selection fitted on train only; test set used once.
R2	Unstable scores: the churn rate changes between snapshots (42% to 69%), so validation and test results differ.	Report Recall, PR-AUC and F1 (not accuracy only); choose the threshold on validation; show results per set.
R3	Churn is a proxy (no order in 90 days), not a real cancellation.	State this clearly and keep the definition fixed across all models.
R4	Segments are not meaningful (for example DBSCAN gives one large cluster plus noise because the data is skewed).	Log transform and scaling; compare with K-Means; choose by Silhouette, Davies-Bouldin and business interpretability.
R5	Library version mismatch breaks the saved models (the preprocessor needs scikit-learn 1.9.0).	Pinned requirements.txt; test loading the models on another machine early.
R6	Integration problems between the app and the models.	The models are finished and the app is under development; the interface (predict_churn, predict_segment) is fixed and tested on sample customers.
R7	Time pressure: design due 6 Nov, implementation due 30 Nov, final 4 Dec.	Prioritize Must requirements; freeze new features on 20 Nov; use sample data for demos.

ID	Risk	Mitigation
R8	A member is unavailable or workload is uneven.	Two members per team as backup; pull-request reviews spread the knowledge.
R9	Cloud deployment is requested late and adds scope.	The platform runs locally; cloud deployment is an optional extension.
R10	Privacy: the app could receive real customer data.	Use only the public anonymized dataset; do not store uploaded files; no secrets in the repository.

4. KPIs (Key Performance Indicators)

Targets are set with the baseline in mind (Precision 0.73, Recall 0.82, F1 0.77, ROC-AUC 0.79, PR-AUC 0.81 on test); the final model should at least match it.

ID	KPI	Target	How it is measured
K1	Recall (churn class)	>= 0.80	Test set, using the threshold chosen on validation.
K2	Precision (churn class)	>= 0.70	Same test set and threshold.
K3	F1-score	>= 0.78	Test set.
K4	ROC-AUC / PR-AUC	>= 0.80 / >= 0.82	Test set.
K5	Silhouette score	>= 0.30	On the chosen model; also report Davies-Bouldin and compare with the other algorithm.
K6	Segments	3 to 6, each >= 5% of customers	Every segment has a business name and description.
K7	Explanation per prediction	100%	Every prediction in the app shows its top factors.
K8	Response time for one customer	<= 2 seconds	Normal laptop, local run.
K9	Batch scoring of 5,253 customers	<= 30 seconds	Upload a CSV and time the full run.
K10	Sample customers scored without errors	20 of 20	Use sample_customers.csv.
K11	Usability	<= 3 minutes	A peer who did not build the app finds a high-risk customer and reads the segment without help (3 to 5 testers).
K12	Reproducibility	100%	A fresh clone runs the notebooks and the app by following the README only.